

example → Comparing values → bar chart

show trends → line chart

show percentage → pie chart

show distribution → histogram

Find relationship → scatter plot

Line plot:

it's used to show trends or changes over time

ex:

Company wants to analyse monthly sales

month	sales
Jan	120
Feb	150
March	180
April	170
May	220

How to plot:

Syntax:

```
import matplotlib.pyplot as plt
```

```
months = ["jan", "feb", "march", "April", "may"]
```

```
sales = [120, 150, 180, 170, 220]
```

```
plt.plot(months, sales)
```

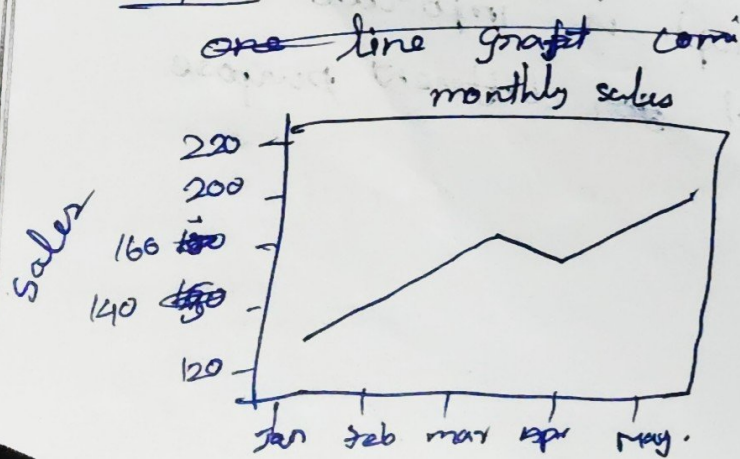
```
plt.title("monthly sales")
```

```
plt.xlabel("months")
```

```
plt.ylabel("sales")
```

```
plt.show()
```

output:



Bar chart:

Compares values between different categories

Student Marks

John 85

Sara 90

David 75

Asha 95

Syntax:

```
students = ["John", "Sara", "David", "Asha"]
```

```
marks = [85, 90, 75, 95]
```

```
plt. plot bar (students, marks)
```

```
plt. title ("students marks")
```

```
plt. x label ("marks")
```

```
plt. show()
```

bar → vertical

barh → horizontal

counting [country]

Output:



pie chart

shows the percentage or proportions

ex:

company	sales
product	sales
laptop	40
mobile	30
Tablet	20
other	10

Syntax:

```
products = ["laptop", "mobile", "laptop", "others"]
```

```
sales = [40, 30, 20, 10]
```

```
plt.pie(sales, labels=products, autopct='%1.1f + %1.1f')
```

```
plt.title("sales report")
```

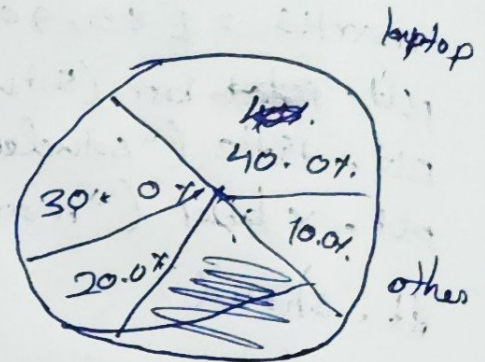
```
plt.show()
```

autopct = '%1.1f + %1.1f'

automatic display
(percentage-wise show())

for election like.

Output



Histogram:

student marks

```
marks = [60, 65, 70, 72, 75, 80, 82, 85, 90, 95]
```

Syntax:

~~plt~~

```
marks = [ ]
```

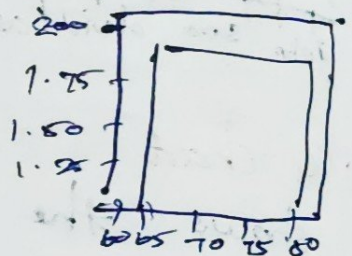
```
plt.hist(marks, bins=5)
```

```
plt.title("marks distribution")
```

```
plt.show()
```

* Frequency range shown

like → age of emp on salary.



Scatter plot:

It shows the relationship between two numerical variables.

```
hours = [2, 4, 5, 6, 8]
```

```
marks = [40, 55, 70, 75, 90]
```

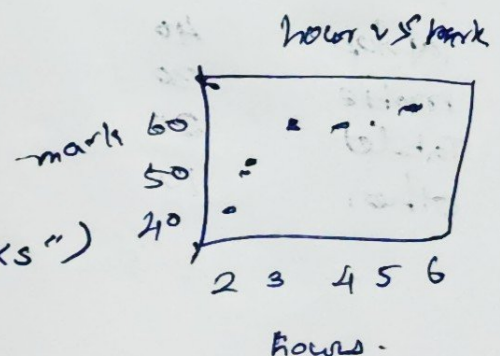
```
plt.scatter(hours, marks)
```

```
plt.title("hours vs marks")
```

```
plt.xlabel("hours")
```

```
plt.ylabel("marks")
```

```
plt.show()
```



autopct = '%.1f%%'

% -> start at format specification

1 -> minimum width

1 -> show 1 digit after the decimal point

f -> floating point numbers.

% -> display the % symbol

1. Create a line graph showing marks of 5 students?

```
import matplotlib.pyplot as plt
```

```
x = ["aa", "bb", "cc", "dd", "ee"]
```

```
y = [90, 66, 72, 88, 98]
```

```
plt.plot(x, y)
```

```
plt.show()
```

2. plot monthly temperature?

```
import matplotlib.pyplot as plt
```

```
x = ["jan", "feb", "mar", "apr", "may"]
```

```
y = [22, 28, 23, 22, 24]
```

```
plt.plot(x, y)
```

```
plt.show()
```

3. Add the x-axis label?

```
import matplotlib.pyplot as plt
```

```
x = ["jan", "feb", "mar", "apr", "may"]
```

```
y = [22, 28, 23, 22, 24]
```

```
plt.plot(x, y)
```

```
plt.xlabel("month")
```

```
plt.show
```

4. Change color, linestyle?

```
import matplotlib.pyplot as plt
```

```
x = ["jan", "feb", "mar", "apr", "may"]
```

```
y = [22, 28, 23, 22, 24]
```

```
plt.plot(x, y, color="red", linestyle="--")
```

```
plt.xlabel("month")
```

```
plt.ylabel("temperature")
```

```
plt.grid()  
plt.show()
```

5. display grid?

```
import matplotlib.pyplot as plt  
x = ["jan", "feb", "mar", "apr", "may"]  
y = [22, 23, 24, 25, 26]
```

```
plt.plot(x, y)
```

```
plt.xlabel("months")
```

```
plt.ylabel("temperature")
```

```
plt.grid()
```

```
plt.show()
```

~~plt.grid()~~